Pure Mathematics 3 Practice Paper (2026 style)

Answers and mark guide

M = method mark. A = accuracy mark; it needs the related M mark. B = independent mark. AG = the answer is given in the question. Accept other correct methods.

Question 1 **[4]**

Square both sides: $9x^2 - 12ax + 4a^2 > 4x^2 + 8ax + 4a^2$ M1

$5x^2 - 20ax > 0$ A1

Critical values $x = 0$ and $x = 4a$ A1

$x < 0$ or $x > 4a$ A1

Question 2 **[5]**

(a) $\int(3e^{2x} - 2e^x)\,dx = \frac{3}{2}e^{2x} - 2e^x$ B1 B1

$\frac{3}{2}e^{2a} - 2e^a - (\frac{3}{2} - 2) = 8$, so $3e^{2a} - 4e^a - 15 = 0$ A1 AG

(b) $(3e^a + 5)(e^a - 3) = 0$ M1

$e^a > 0$, so $e^a = 3$ and $a = \ln 3$ A1

Question 3 **[6]**

(a) $(p + q)^3 - 3pq(p + q) = p^3 + 3p^2q + 3pq^2 + q^3 - 3p^2q - 3pq^2 = p^3 + q^3$ B1 AG

(b) Put $p = \sin^2 x$ and $q = \cos^2 x$, so $p + q = 1$ M1

$\sin^6 x + \cos^6 x = 1 - 3\sin^2 x\cos^2 x$ A1

Use $\sin 2x = 2\sin x\cos x$: $\sin^6 x + \cos^6 x = 1 - \frac{3}{4}\sin^2 2x$ M1

$1 - \frac{3}{4}\sin^2 2x = \frac{7}{16}$ gives $\sin^2 2x = \frac{3}{4}$, so $\sin 2x = \pm\frac{1}{2}\sqrt{3}$ A1

$x = \frac{1}{6}\pi$, $\frac{1}{3}\pi$, $\frac{2}{3}\pi$, $\frac{5}{6}\pi$ (all four, and no others) A1

Question 4 **[8]**

(a) $\frac{dx}{d\theta} = -4\sin\theta$ and $\frac{dy}{d\theta} = 3\cos\theta$ B1

$\frac{dy}{dx} = -\frac{3\cos\theta}{4\sin\theta}$ M1

$y - 3\sin\theta = -\frac{3\cos\theta}{4\sin\theta}(x - 4\cos\theta)$ M1

$4y\sin\theta - 12\sin^2\theta = -3x\cos\theta + 12\cos^2\theta$, so $3x\cos\theta + 4y\sin\theta = 12$ A1 AG

(b) $A = (\frac{4}{\cos\theta}, 0)$ and $B = (0, \frac{3}{\sin\theta})$ B1

Area $= \frac{1}{2} \times \frac{4}{\cos\theta} \times \frac{3}{\sin\theta} = \frac{6}{\sin\theta\cos\theta} = \frac{12}{\sin 2\theta}$ M1

$\frac{12}{\sin 2\theta} = 16$, so $\sin 2\theta = \frac{3}{4}$ A1

$\theta = 0.424$ or $\theta = 1.15$ A1

Question 5 **[8]**

(a) $r = \frac{1}{2}\sqrt{2}$ B1

$\theta = \frac{1}{4}\pi$ B1

(b) $P_nP_{n+1} = |w^{n+1} - w^n| = |w|^n|w - 1|$ M1

$|w - 1| = |-\frac{1}{2} + \frac{1}{2}i| = \frac{1}{2}\sqrt{2}$, so $P_nP_{n+1} = (\frac{1}{2}\sqrt{2})^{n+1}$. Each length is $\frac{1}{2}\sqrt{2}$ times the length before it, so the lengths form a GP. A1

Total length $= \frac{\frac{1}{2}\sqrt{2}}{1 - \frac{1}{2}\sqrt{2}}$ (sum to infinity, as $|r| < 1$) M1

$= \frac{1}{\sqrt{2} - 1} = 1 + \sqrt{2}$ A1

(c) $OP_n = (\frac{1}{2}\sqrt{2})^n < 0.01$, so $n > \frac{\ln 0.01}{\ln(\frac{1}{2}\sqrt{2})} = 13.3$ M1

$n = 14$ A1

Question 6 **[8]**

(a) Area of water $= \frac{1}{2} \times 1^2 \times (\theta - \sin\theta)$ B1

The tank has a constant cross-section, so $\frac{1}{2}(\theta - \sin\theta) = \frac{1}{4}\pi$. This gives $\theta = \frac{1}{2}\pi + \sin\theta$. B1 AG

(b) Let $f(\theta) = \theta - \sin\theta - \frac{1}{2}\pi$: $f(2.2) = -0.179$ and $f(2.4) = 0.154$ M1

The sign changes and f is continuous, so the root is between 2.2 and 2.4 A1

(c) Use $\theta_{n+1} = \frac{1}{2}\pi + \sin\theta_n$ correctly at least once M1

For example: 2.3, 2.3165, 2.3054, 2.3129, 2.3079, 2.3112, 2.3090, 2.3105, 2.3095, 2.3102. So $\theta = 2.31$ (to 2 d.p.). A1

Depth $= 1 - \cos(\frac{1}{2}\theta)$ M1

Depth $= 0.596$ m A1

Question 7 **[9]**

(a) $\frac{dy}{dx} = e^{-\frac{1}{2}x} - \frac{1}{2}xe^{-\frac{1}{2}x}$ M1 A1

$\frac{dy}{dx} = 0$ gives $x = 2$, so $M = (2,\ 2e^{-1})$ A1

(b) $\int_0^2 xe^{-\frac{1}{2}x}\,dx = [-2xe^{-\frac{1}{2}x}]_0^2 + \int_0^2 2e^{-\frac{1}{2}x}\,dx$ (by parts) M1 A1

$= -4e^{-1} + [-4e^{-\frac{1}{2}x}]_0^2 = 4 - 8e^{-1}$ A1

Area below OM (a triangle) $= \frac{1}{2} \times 2 \times 2e^{-1} = 2e^{-1}$ B1

Area above $OM = 4 - 8e^{-1} - 2e^{-1} = 4 - 10e^{-1}$ M1

Ratio $(4 - 10e^{-1}) : 2e^{-1} = (4e - 10) : 2 = (2e - 5) : 1$ A1 AG

Note: OM is $y = x/e$, and $xe^{-\frac{1}{2}x} \geqslant x/e$ for $0 \leqslant x \leqslant 2$, so the line is below the curve.

Question 8 **[9]**

(a) $\frac{1}{(N-200)(800-N)} = \frac{1}{600(N-200)} + \frac{1}{600(800-N)}$ B1 B1

(b) Separate the variables: $\int \frac{dN}{(N-200)(800-N)} = \int \frac{dt}{5000}$ B1

$\frac{1}{600}\ln(N-200) - \frac{1}{600}\ln(800-N) = \frac{1}{5000}t + c$ M1 A1

$t = 0,\ N = 500$ gives $c = 0$ M1

$\ln\frac{N-200}{800-N} = 0.12t$, so $N = \frac{800e^{0.12t} + 200}{e^{0.12t} + 1}$ A1

(c) The number of fish tends to 800 B1

When $0 < N < 200$, $\frac{dN}{dt} < 0$, so N decreases: the fish in the second lake die out B1

Question 9 **[9]**

(a) Direction $\overrightarrow{AB} = -3\mathbf{i} - \mathbf{j} - 2\mathbf{k}$ B1

$\mathbf{r} = 5\mathbf{i} + 3\mathbf{j} + 6\mathbf{k} + \mu(3\mathbf{i} + \mathbf{j} + 2\mathbf{k})$ B1

(b) $1 + 2\lambda = 5 + 3\mu,\ -\lambda = 3 + \mu,\ 4 + 2\lambda = 6 + 2\mu$ M1

Two of the equations give $\lambda = -1$ and $\mu = -2$ A1

The third equation is true ($2 = 2$), so the lines meet at C: $-\mathbf{i} + \mathbf{j} + 2\mathbf{k}$ A1

(c) $D = (1 + 2\lambda)\mathbf{i} - \lambda\mathbf{j} + (4 + 2\lambda)\mathbf{k}$ and $\overrightarrow{AD} \cdot (2\mathbf{i} - \mathbf{j} + 2\mathbf{k}) = 0$ give $9\lambda - 9 = 0$ M1

$\lambda = 1$, so D is $3\mathbf{i} - \mathbf{j} + 6\mathbf{k}$ A1

$AD = 2\sqrt{5}$ and $CD = 6$, and angle $ADC = 90°$ M1

Area $= \frac{1}{2} \times 2\sqrt{5} \times 6 = 6\sqrt{5}$ A1

Question 10 **[9]**

(a) $x^2 + y^2 = 4$ and $x^2 + (y - 2)^2 = 4$ M1

Subtract: $y = 1$ A1

P is $\sqrt{3} + \mathrm{i}$ and Q is $-\sqrt{3} + \mathrm{i}$ A1

(b) The greatest value of $\arg z$ is at Q M1

$\arg z = \frac{5}{6}\pi$ A1

(c) Chord PQ makes an angle of $\frac{2}{3}\pi$ at each centre B1

Each segment: $\frac{1}{2} \times 2^2 \times (\frac{2}{3}\pi - \sin\frac{2}{3}\pi) = \frac{4}{3}\pi - \sqrt{3}$ M1 A1

Area of $R = \pi \times 2^2 - 2(\frac{4}{3}\pi - \sqrt{3}) = \frac{4}{3}\pi + 2\sqrt{3}$ A1